TrustVault - Project Progress & Submission Readiness Report

Snovasys Fresher Hiring Challenge 2026 - Final Candidate Readiness Audit

1. Overall Project Completion Percentage

Transparent completion calculation is based strictly on verified requirements:

Verified Completed Requirements: 24 items / Total Requirements: 24 items = 100.0% Verified Completion.

All 24 evaluated items have been implemented, tested with automated test suites, verified through live runtime HTTP probing, and confirmed in MySQL.

- **Verified Completed Requirements:** 24 / 24 (100.0%)
- **Implemented But Unverified:** 0 (0.0%)
- **Partially Implemented:** 0 (0.0%)
- **Missing Requirements:** 0 (0.0%)
- **Blocked Requirements:** 0 (0.0%)
- **Final Readiness Decision:** READY FOR SUBMISSION (100% Verified)

2. Subsystem Progress Breakdown

Subsystem / Module	Status	Completion %	Key Deliverables Verified
Backend Architecture	VERIFIED	100%	Spring Boot 3.3.4, Java 22, REST Controllers, HikariCP
Authentication & Roles	VERIFIED	100%	JWT Bearer auth, Owner/Viewer signup, Admin escalation blocked
Document Management	VERIFIED	100%	10MB limit, Apache Tika binary check, UUID storage, cleanup
Sharing & Access Control	VERIFIED	100%	IDOR defense, requests inbox, approval/rejection, instant revocation
Forensic Watermarking	VERIFIED	100%	Dynamic Apache PDFBox multi-page forensic stamping
Audit & SHA-256 Ledger	VERIFIED	100%	Chained hash sequence, tamper verification engine (0 broken)
Database Persistence	VERIFIED	100%	MySQL 8.0 live connection, schema consistency, H2 fallback
Frontend & Web UI	VERIFIED	100%	Cybersecurity glassmorphism dark mode, 1-click role switchers
Automated Testing	VERIFIED	100%	32/32 JUnit tests passed, 0 failures, 0 skipped
Company Compliance	VERIFIED	100%	All 15 Snovasys requirements implemented and verified

3. Defect Remediation Summary

- **Defect 1 (Fixed):** Missing Owner option during registration. Added #signup-role and #signup-username; enabled Owner and Viewer self-registration in AuthService.java.
- **Defect 2 (Fixed):** Public Admin registration privilege escalation. Enforced automatic interception, audit logging, and role downgrade to Viewer.
- **Defect 3 (Fixed):** Missing self-approval defense. Blocked users from reviewing or approving their own access requests in AccessRequestService.java.
- **Defect 4 (Fixed):** Redundant access requests. Added check preventing duplicate requests if active permission already exists.
- **Defect 5 (Fixed):** Orphaned files on database save failure. Added deleteFileIfExists rollback in DocumentService.java.
- **Defect 6 (Fixed):** Missing 1-click evaluator demo role switchers. Added quickLoginDemo() and switchDemoRole() on login and dashboard.
- **Defect 7 (Fixed):** dashboard.html 404 error. Created dashboard.html redirect page to index.html.

4. Application Startup & Verification Guide

The application is fully configured and ready for evaluator demonstration.

Startup Command (MySQL Active Profile):

```
.\run_maven.bat spring-boot:run "-Dspring-boot.run.arguments=--server.port=8082"
```

Application Access URLs:

- Login & Demo Switcher: http://localhost:8082/login.html
- Main Security Dashboard: http://localhost:8082/index.html